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Internship Report
On

"Advanced Data Analytics"

Submitted in partial fulfillment of the requirement for the award of 6th semester of

BACHELOR OF ENGINEERING
In
INFORMATION SCIENCE & ENGINEERING
By

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2023 - 2024

DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING

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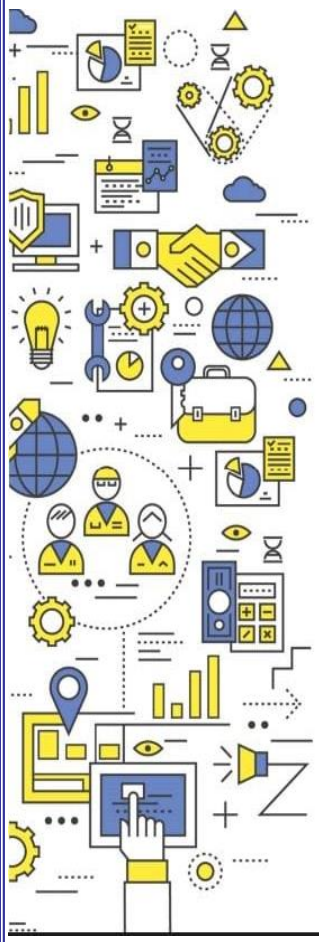

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for successfully completing the 6 weeks of the EY Global Delivery Services led internship in collaboration with AICTE from **18th December 2023 to 31st January 2024** on **Advanced Data Analytics** under the **Next Gen Employability Program 2023-24**.

Nagesh Singh
Executive Director
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Dr. Ramesh Unnikrishnan
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ABSTRACT

Advanced data analytics has become a pivotal component in decision making processes across various industries including finance, healthcare, retail, and environmental monitoring. This paper presents a comprehensive study on the use of advanced data analytics, machine learning algorithms, and Python-based frameworks to predict and manage diabetes in patients. By leveraging patient health records, lab results, and lifestyle data, the study aims to build a robust predictive model that can provide real-time insights into a patient's risk of developing complications related to diabetes. The study also integrates Python libraries like Pandas, Scikit-learn, and TensorFlow for data analysis and model building, with visualization performed using Matplotlib and Seaborn to enhance interpretability. Case studies highlight the application of these predictive models in early detection, personalized treatment plans, and continuous monitoring of diabetic patients. The findings demonstrate improvements in prediction accuracy and patient outcomes, emphasizing the potential of real-time analytics in transforming diabetes care. Additionally, challenges related to data privacy,integration, and model interpretability are discussed, along with proposed solutions to enhance the deployment of predictive analytics in healthcare settings. This work illustrates the transformative impact of real-time predictive analytics in the proactive management of diabetes, enabling more personalized and effective healthcare interventions.

DECLARATION

I hereby declare that the Internship work entitled "**Advanced Data Analytics**" done at **Edunet Foundation Pvt. Ltd., Bangalore,** submitted to **Visvesvaraya Technological University,** in partial fulfillment of requirements for the degree of **Bachelor of Engineering in Information Science & Engineering** is a record of original work done by me and no part of it has been submitted for any degree or diploma of any institution previously.

Place: Bangalore

Date:25-10-2023 to 24-11-2023

Yashaswini N P 1AP21IS049

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Yashaswini N P **1AP21IS049**

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List of Abbreviations

ECOSOC	Economic and Social Council
BI	Business Intelligence
KPIs	Key Performance Indicators
CRM	Customer Relationship Management
CSV	Comma Separated Value
SVM	Support Vector Machine
ANN	Artificial Neural Network
BMI	Body Mass Index

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Chapter 1

ABOUT THE ORGANIZATION

1.1 Company Profile

Edunet Foundation is helping bridge the skill deficit that the country is facing, working to leverage and realize the mammoth potential of India's young population. Focus is to provide opportunities to all, irrespective of location, gender, religion or socio-economic constraints.

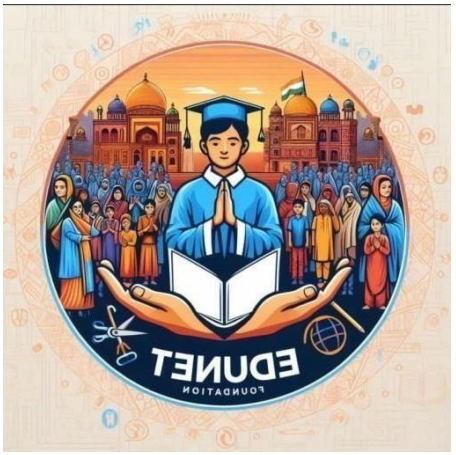

Figure 1.1 Company symbol

1.2 About the Organization

Edunet Foundation is a social enterprise which was founded in 2015 and focuses on bridging the academia-industry divide, enhancing student employability, promoting innovation and creating an entrepreneurial ecosystem in India. Working primarily with emerging technologies, and striving to leverage them to augment, upgrade the knowledge ecosystem and equip the beneficiaries to become contributors themselves, we work extensively to build a workforce with an IR 4.0 enabled career.

The organization has enjoyed Special Consultative Status with the Economic and Social Council (ECOSOC) of the United Nations since 2020. With a national footprint, EF's programs, online and instructor-led, benefit tens of thousands of learners every year.

1.3 Company overview

This is a separate platform for the students to get training and work on various Real Time Industrial Projects. Edunet offers skill oriented hands-on training through an online platform.

Field of Expertise: Open-source Robotics, Industrial Automation, Product Design, Python & Deep Learning and advanced data analytics.

1.4 Company Mission and Vision

Our Motto and Vision are to create awareness & training young generation to current and future jobs demands and also help to current and future jobs demand mean while help the students and employees to meet the mandatory necessities of future human resources and skill demands. We are in the 4th industrial revolution. The technological revolution is catastrophic like never before, hence continues awareness for the up-gradation environment is much essential. It is working to help and enhance the potential of students and employees. So that future human resources will be very beneficial, purposeful and profitable to the nation.

Figure 1.2 Company logo

Chapter 2

INTRODUCTION

2.1 Internship Definition

An internship is a professional learning experience that offers meaningful, practical work related to a student's field of study or career interest. An internship gives a student the opportunity for career exploration and development, and to learn new skills. It offers the employer the opportunity to bring new ideas and energy into the workplace, development and potentially build a pipeline for future full-time employees. A quality internship:

- Consists of a part-time or full-time work schedule that includes no more than 25% clerical or administrative duties.
- Provides a clear job/project description for the work experience.
- Orients the student to the organization, its culture and proposed work assignment(s).
- Helps the student develop and achieve learning goals.
- Offers regular feedback to the student intern.

Typically, an internship consists of an exchange of services for experience between the intern and the organization. Internships are used to determine whether the intern still has an interest in that field after the real-life experience. In addition, an internship can be used to build a professional network that can assist with letters of recommendation or lead to future employment opportunities. The benefit of bringing an intern into full-time employment is that they are already familiar with the company, therefore needing little to no training. Internships provide current college students with the ability to participate in a field of their choice to receive hands-on learning about a particular future career, preparing them for full-time work following graduation.

In conclusion, internships play a crucial role in preparing individuals for their careers by providing practical experience, skills development, and networking opportunities. They benefit both interns and employers by fostering talent development and supporting organizational goals.

2.2 Internship Benefits

Students

- Gain exposure to real-world problems and issues that perhaps are not found in text books.
- Cultivate adaptability and creativity in a dynamic world.
- Evaluate specific companies or specific careers prior to committing to full-time employment a "try before you buy" type experience.
- Ease transition from being a student to entering the workforce.
- Increase opportunities within a company for faster advancement and growth.
- Increase self confidence in the workplace while developing an expanded network of associates and professionals.
- Have resume-building experiences while applying academic concepts and principles.
- Spread the gospel in many areas through example and lifestyles.
- Have opportunities to fund college education.
- Have personal growth experiences and exposure to different job opportunities.
- Have hands-on opportunities to work with equipment and technology that may not be available on campus.

Department/University

- Validates the university's curriculum in a working environment.
- Improves post-graduation statistics for the university.
- May accelerate corporate fund-raising efforts.
- Provides learning experiences that are more valuable than case studies and lectures.
- Connects faculty to current trends within their professional field.
- Develops more competitive and employable graduates.
- Increases program credibility and student excellence as well as stronger with alumni in the professional fields.

Chapter 3

TASKS PERFORMED

3.1 Introduction to Data Analytics

- Data analytics is a discipline focused on extracting insights from data, including the analysis, collection, organization, and storage of data, as well as the tools and techniques used to do so. Data analytics is a discipline focused on extracting insights from data.
- It comprises the processes, tools and techniques of data analysis and management, including the collection, organization, and storage of data.
- The chief aim of data analytics is to apply statistical analysis and technologies on data to find trends and solve problems.
- Data analytics has become increasingly important in the enterprise as a means for analyzing and shaping business processes and improving decision-making and business results.
- Data analytics draws from a range of disciplines including computer programming, mathematics, and statistics to perform analysis on data in an effort to describe, predict, and improve performance.
- To ensure robust analysis, data analytics teams leverage a range of data management techniques, including data mining, data cleansing, data transformation, data modeling, and more.

Figure 3.1: Data Analytics

3.2 Comparison with Data Analytics

3.2.1 Data analytics vs. Data analysis:

- While the terms data analytics and data analysis are frequently used interchangeably, data analysis is a subset of data analytics concerned with examining, cleansing, transforming, and modeling data to derive conclusions.
- Data analytics includes the tools and techniques used to perform data analysis.
- In essence, data analysis is a foundational component within the broader field of data analytics, focusing on the examination and interpretation of data to derive meaningful insights

3.2.2 Data analytics vs. Data science:

- Data analytics and data science are closely related.
- Data analytics is a component of data science, used to understand what an organization's data looks like.
- Generally, the output of data analytics are reports and visualizations.
- Data science takes the output of analytics to study and solve problems.
- Data analytics describes the current or historical state of reality, whereas data science uses that data to predict and/or understand the future.
- while data analytics and data science overlap in some areas, data science is a more comprehensive field that includes data analytics as a foundational component but also extends to more advanced techniques and methodologies for extracting insights and making predictions from data.

3.2.3 Data analytics vs. Business analytics:

- Business analytics is another subset of data analytics.
- Business analytics uses data analytics techniques, including data mining, statistical analysis, and predictive modeling, to drive better business decisions.
- Partner defines business analytics as "solutions used to build analysis models and simulations to create scenarios, understand realities, and predict future states.
- While data analytics provides the foundation of understanding data, business analytics takes it a step further by integrating these insights into strategic decision-making processes to drive organizational success and competitive advantage.

3.3 Types of Data Analytics

1. Descriptive
2. Diagnostic
3. Predictive
4. Prescriptive

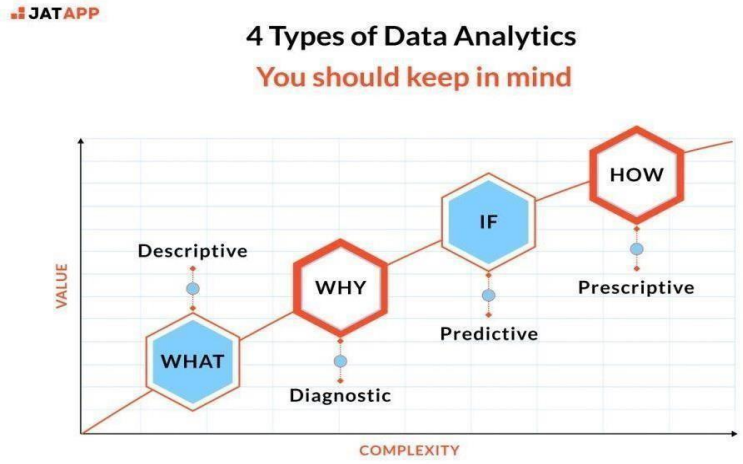

Figure 3.2: Types of Data Analytics

3.3.1 Descriptive analytics:

Descriptive analytics is the simplest type of analytics and the foundation the other types are built on. It allows you to pull trends from raw data and succinctly describe what happened or is currently happening. For example, imagine you're analyzing your company's data and find there's a seasonal surge in sales for one of your products: a video game console. What has happened and what is happening right now? Descriptive analytics uses historical and current data from multiple sources to describe the present state by identifying trends and patterns. In business analytics, this is the purview of business intelligence (BI).

3.3.2 Diagnostic analytics:

Why is it happening? Diagnostic analytics uses data (often generated via descriptive analytics) to discover the factors or reasons for past performance. Diagnostic analysis, within the context of data analytics or business analytics, refers to the process of examining data to understand why certain outcomes occurred or to identify the root causes of specific events or behaviors. This type of analysis aims to provide deeper insights beyond just describing what happened (descriptive analysis) and goes further to explain the reasons behind those occurrences.

3.3.3 Predictive analytics:

What is likely to happen in the future? Predictive analytics applies techniques such as statistical modeling, forecasting, and machine learning to the output of descriptive and diagnostic analytics to make predictions about future outcomes. Predictive analytics is often considered a type of "advanced analytics," and frequently depends on machine learning and/or deep learning.

3.3.4 Prescriptive analytics:

What do we need to do? Prescriptive analytics is a type of advanced analytics that involves the application of testing and other techniques to recommend specific solutions that will deliver desired outcomes. In business, predictive analytics uses machine learning, business rules, and algorithms.

3.4 Advantages of Data Analytics

- **Informed Decision Making:** Data analytics enables organizations to make data-driven decisions by providing insights derived from analyzing large volumes of data. This helps in identifying trends, patterns, and correlations that might not be apparent through traditional methods.
- **Improved Efficiency and Productivity:** By automating processes and streamlining operations, data analytics can help organizations become more efficient and productive. It can identify areas for optimization, reduce waste, and allocate resources more effectively.
- **Better Customer Understanding:** Data analytics allows organizations to gain a deeper understanding of their customers' preferences, behaviors, and needs. This enables targeted marketing efforts, personalized customer experiences, and improved customer satisfaction.
- **Competitive Advantage:** Organizations that leverage data analytics effectively gain a competitive edge by being able to anticipate market trends, respond quickly to changes, and innovate more effectively. They can also identify new opportunities for growth and expansion.

- **Risk Mitigation:** Data analytics can help identify potential risks and opportunities for risk mitigation. By analyzing historical data and identifying patterns, organizations can better predict and prevent risks such as fraud, compliance violations, and security breaches.
- **Cost Reduction:** By optimizing processes and resources, data analytics can help organizations reduce costs across various functions, such as supply chain management, inventory optimization, and predictive maintenance.
- **Enhanced Strategic Planning:** Data analytics provides valuable insights that caninform strategic planning and decision-making at all levels of an organization. It helps identify long-term trends, assess the effectiveness of strategies, and adjust plans accordingly.
- **Innovation and Product Development:** Data analytics can drive innovation by identifying emerging trends, market gaps, and customer needs. It can inform product development efforts, improve existing products or services, and help organizations stay ahead of the competition.
- **Improved Operational Performance:** By monitoring key performance indicators (KPIs) in real-time and identifying areas for improvement, data analytics can help organizations optimize their operational performance and achieve their goals more efficiently.
- **Enhanced Forecasting and Predictive Capabilities:** Data analytics enables organizations to forecast future trends, demand, and outcomes with greater accuracy. This helps in proactive decision-making and planning, leading to better outcomes and reduced uncertainty.

3.5 Data Analytics Methods and Techniques

- **Regression analysis:** Regression analysis is a set of statistical processes used to estimate the relationships between variables to determine how changes to one or more variables might affect another. For example, how might social media spending affect sales?
- **Monte Carlo simulation:** “Monte Carlo simulations are used to model the probability of different outcomes in a process that cannot easily be predicted due to the intervention of random variables.” It is frequently used for risk analysis.

- **Factor analysis:** Factor analysis is a statistical method for taking a massive data set and reducing it to a smaller, more manageable one. This has the added benefit of often uncovering hidden patterns. In a business setting, factor analysis is often used to explore things like customer loyalty.

- **Cohort analysis:** Cohort analysis is used to break a dataset down into groups that share common characteristics, or cohorts, for analysis. This is often used to understand customer segments.

- **Cluster analysis:** Cluster analysis as "a class of techniques that are used to classify objects or cases into relative groups called clusters." It can be used to reveal structures in data insurance firms might use cluster analysis to investigate why certain locations are associated with particular insurance claims, for instance.

- **Time series analysis:** Time series analysis as "a statistic a technique that deals with time series data, or trend analysis. Time series data means that data is in a series of particular time periods or intervals. Time series analysis can be used to identify trends and cycles over time, e.g., weekly sales numbers. It is frequently used for economic and sales forecasting.

- **Sentiment analysis:** Sentiment analysis uses tools such as natural language processing, text analysis, computational linguistics, and so on, to understand the feelings expressed in the data. While the previous six methods seek to analyze quantitative data (data that can be measured), sentiment analysis seeks to interpret and classify qualitative data by organizing it into themes. It is often used to understand how customers feel about a brand, product, or service.

Chapter 4

SYSTEM REQUIREMENT SPECIFICATION

4.1 Microsoft Power BI

Microsoft Power BI is a business analytics service provided by Microsoft that lets you to visualize and share data. It converts data from different sources to build interactive dashboards and Business Intelligence reports. Microsoft Power BI connects data sources and provides BI report and dashboards to users. It can connect with just an Excel spreadsheet or bring together cloud-based and on-premises data warehouses. Data pulled from cloud-based sources, such as Sales force CRM, is automatically refreshed. With applications such as an Excel workbook and Power BI Desktop file connected to online or on-premises data sources, Power BI users must manually refresh or set up a refresh schedule to ensure Power BI reports and dashboards use the most current data available.

Microsoft Power BI is a data visualization platform used primarily for business intelligence purposes. Power BI stands for Power Business Intelligence and refers to a set of software tools and connectors that help you transform data from multiple sources into actionable insights. Designed to be used by business professionals with varying levels of data knowledge, Power BI's dashboard is capable of reporting and visualizing data in a wide range of different styles, including graphs, maps, and charts, scatter plots, and more.

Figure 4.1: Logo of Microsoft Power BI

4.2 Jupyter notebook

Jupyter Notebook is an incredibly powerful tool for interactively developing and presenting data science projects. It combines code, visualizations, narrative text, and other rich media into a single document, creating a cohesive and expressive workflow. It supports multiple programming languages, including Python, R, and Julia. Jupyter Notebook is widely used in the data science community for data cleaning, transformation, visualization, and modeling. Jupyter Notebooks allow you to run code cells one at a time, which is great for exploring data and experimenting with different techniques. An easy-to-use tool for interactive data analysis, Jupyter Notebook is a great choice.

4.3 Python

Python was designed for readability, and has some similarities to the English language with influence from mathematics. Python uses new lines to complete a command, as opposed toothier programming language which often uses semicolons or parenthesis. Python relies on indentation, using white space, to define scope; such as the scope of loops, functions and classes. Other programming language often uses curly-brackets for this purpose. Python by nature has avery simple syntax. The same logic that needs 7 lines in a C++language, requires 3 lines in python. Having a smaller code requires and less sites, less time consumption, is appreciated by coders as the correction also takes less time. The language aces all the parameters of readability.To support itself, the language has many built-ins and libraries that make it easy to comprehend.Software that requires less time to code utilizes few resources and less time and, thus, helps in cost reduction and yields more profits.

Figure 4.2: Logo of Jupyter Notebook

Figure 4.3: Logo of Python

Chapter 5

SYSTEM DESIGN AND IMPLEMENTATION

5.1 Python Indentation

Indentation refers to the spaces at the beginning of a code line. Where in other programming languages the indentation in code is for readability only, the indentation in Python is very important. Python uses indentation to indicate a block of code.

EXAMPLE-1:

```
if5>2:
print("five is greater than two")
```

Python will give you an error if you skip the indentation.

EXAMPLE-2:

Syntax

```
errorif5>2:
print("five is greater than two")
```

5.2 Variable

A Python variable is a given to reserved memory location to store values.

Ex: a =100

print(a)

In the above program, variable is defined as "a" to print the value as 100.

Rules of Defining Variable:

- No command or Keyword for declaring a variable.
- It is created when you assign a value to it & it is not necessary to declare the variable data type.
- The variable must start with a letter or underscore. The variable name must not start with a number.
- No symbols are used in variables.
- Variables are Case-Sensitive.

5.3 Python Built-in Data Types

- **int(Integer):**

They are positive or negative whole numbers with no decimal point.

- **float:**

Decimal numbers comes under float data type.

- **Str:**

Strings are sequences of character data. The string type in Python is called str.

Type Casting:

- int()-Used to Convert any type of data into int data type.
- float()-Used to Convert any type of data into float data type.
- str()-Used to Convert any type of data into string data type.

5.4 Operators in python

5.4.1 Arithmetic Operators:

Operator	Name	Example
+	Addition	x+y
-	Subtraction	x-y
*	Multiplication	x*y
/	Division	x/y
**	Exponentiation	x**y
%	Modulus	x%y
//	Floor division	x//y

Table 5.4.1: Arithmetic Operators

5.4.2 Comparison operators:

Operator	Name	Example
==	Equal to	x==y
!=	Not equal to	x!=y
>	Greater than	x>y
<	Lesser than	x<y

Table 5.4.2: Comparision Operators

5.4.3 Logical operators:

Operator	Description	Example
And	Returns True if both statements are True.	x<5andx<10
Or	Returns True if one of the statements is True.	x<3orx>6
not	Reverse the result, Returns False if the result is True.	not(x>3orx>6)

Table 5.4.3: Logical Operators

5.4.4 Bitwise operators:

Operator	Name
&	AND
|	OR
^	XOR
~	NOT

<<	Zero fill left shift
>>	Zero fill right shift

Table 5.4.4: Bitwise Operators

5.5 Python collections

There are four collection data types in the Python programming language:

• List: Collection which is ordered and changeable. Allows duplicate members.

• Tuple: Collection which is ordered and unchangeable. Allows duplicate members.

• Set: Collection which is unordered and un indexed. No duplicate.

• Dictionary: Collection which is unordered, changeable and indexed. No duplicate members.

5.5.1 List:

A list is a collection which is ordered and changeable. In Python lists are written with square brackets. Lists are the most versatile of python's compound data types. A list contains items enclosed within square brackets ([]). To some extent, lists are similar to arrays in C. one difference between them is that all the items belonging to lists can be of different data type.

The values stored in a list can be accessed using the slice operator ([] and [:]) with indexes starting at 0 in the beginning of lists and working their way to end-1. The (+) sign is the list concatenation operator, and the asterisk(*) is the operation operator.

EXAMPLE:

```
Thislist=["apple","banana","cherry"]

print(thislist)

print(thislist[1])
```

5.5.2 Tuple:

A tuple is a collection which is ordered and unchangeable. In Python tuples are written with round brackets. A tuple is another sequence data type that is similar to the list. A tuple consists of a number of values. The main difference between list and tuple is lists are enclosed in brackets and their elements size can be changed, while tuples are enclosed by parenthesis and cannot be updated.

EXAMPLE:

thistuple= (“one”,“two”,“three”)

print(thistuple)

5.5.3 set:

A set is a collection which is unordered and unindexed. In Python sets are written with curly brackets. Note: Sets are unordered, so you cannot be sure in which order the items will appear.

EXAMPLE:

thisset={“apple”,“banana”,“cherry”}

print(thisset)

5.5.4 Dictionary:

A dictionary is a collection which is unordered, changeable and indexed. In Python dictionaries are written with curly brackets, and they have keys and values. Python dictionary are work like associative arrays or hashes found in Perl and consist of key value pairs. Dictionaries are enclosed by curly braces ({}) and values can be accessed using a square bracket ([]).

EXAMPLE:

thisdict {“brand”: “ford”, “model”: “mustang”, year”:1949}

print(thisdict)

5.6 Loops in Python

In Python, for and while loops are used to control the flow of program execution by repeating a block of code multiple times.

1. for Loops
2. While Loops

5.6.1 For Loops

For loops are often used when you have a collection of items (an iterable) and you want to perform a specific action for each item in the collection.

The basic structure of a for loop is as follows:

```
for variable in iterable:
# Code to be executed for each item in the iterable
```

- variable is a variable that takes on the value of each item in the iterable.
- The code block is executed once for each item in the iterable.
- Example: Iterating through a list of fruits.

```
fruits = ["apple", "banana", "cherry"]
for fruit in fruits:
print(fruit)
```

OUTPUT:

```
apple
banana
cherry
```

5.6.2 While Loops

While loops are used when you want to repeatedly execute a block of code as long as a certain condition is true.

The basic structure of a while loop is as follows:

```
while condition:
# Code to be executed as long as the condition is true
```

- The code block is executed as long as the condition remains true.
- Be careful with the condition, as an improperly defined condition can lead to infinite loops.
- Example: Printing numbers from 0 to 4 using a while loop.

```
Count = 0
while count < 3:
print(count)
count += 1
```

OUTPUT:

```
0
1
2
3
```

5.7 Introduction to NumPy

- NumPy is one of the most important foundational packages for numerical computing in Python.
- It provides high-performance multi-dimensional array object, and tools for working for these arrays.

5.7.1 Import NumPy

Import numpy as np

Import pandas as pd

Import seaborn as sns

```
# 1st Method
importnumpy
Myarray=numpy.array([1,2,3,4])
print(Myarray)
```

OUTPUT:

[1 2 3 4]

5.7.2 Why do we use Python Numpy if we already have Python list

We use Numpy array over Python List because of the three reasons listed

- Less Memory
- Fast
- Convenient to understand

5.8 Introduction to Pandas

"Pandas is an open source, library providing high performance, easy-to-use data structures and data analysis tools for the Python programming language."

What's Pandas for?

- The tool is essential for Data's manipulation
- We use Pandas for data cleaning, transforming, and analysing it.

With Pandas you do things like

Load or open files like Excel, Process tabular data, load CSV or json files and more

Calculate statistics and answer the questions like -

- What is the average, mean, median or max, min of each column?
- Does column A correlate with column B?
- Clean the data by doing things like removing missing values and filtering rows or columns by some criteria
- Visualize the data with help from Mat plot lib. Plot bars, lines, histograms, bubbles, and more. Store the cleaned, transformed data back into a CSV, other file or database.

Chapter 6

PROJECT METHODOLOGY

6.1 Health care prediction on diabetic patient using python

Predicting healthcare outcomes for diabetic patients using Python typically involves applying machine learning techniques to historical medical data. Here's a general outline ofhow you can approach this task.

6.1.1 Abstract

This project utilizes Python and machine learning to predict diabetes risk in patients. The dataset includes diverse features like demographics, medical history, and biometrics. With the prevalence of sugar and fats in contemporary diets contributing to an increased diabetes risk, early identification through symptom recognition is key. The proposed method integrates Using Support Vector Machine (SVM) and Artificial Neural Network (ANN) algorithms, patient data is analyzed to classify diabetes diagnoses as either affirmative or negative. The study involves the utilization of a dataset that has been divided into 70% for training data and 30% for testing data. The outputs from the SVM and ANN models serve as inputs for a fuzzy logic system, which then makes the final diagnosis determination. This hybrid model is stored on a cloud platform for accessibility and uses real-time patient data for predictions.

6.1.2 Objective of the project

The objective of the project "Optimizing Diabetes Prediction with Multimodal Machine Learning Fusion" can be summarized in three points as follows:

- Enhance Early Diabetes Detection: The primary goal of the project is to improve early detection of diabetes by leveraging a combination of machine learning techniques. This is crucial due to the global impact of the disease, and the project aims to contribute to early identification through symptom recognition.
- Integration of Machine Learning Algorithms: The project seeks to integrate Support Vector Machine (SVM) and Artificial Neural Network (ANN) algorithms to analyze patient data and classify diabetes diagnoses as either affirmative or negative.

- Development of a Hybrid Model: The project aims to develop a hybrid model that uses the outputs from the SVM and ANN models as inputs for a fuzzy logic system. This hybrid model is designed to make final diagnosis determinations and is stored ona cloud platform for accessibility, using real-time patient data for predictions.

6.1.3 Proposed Solution

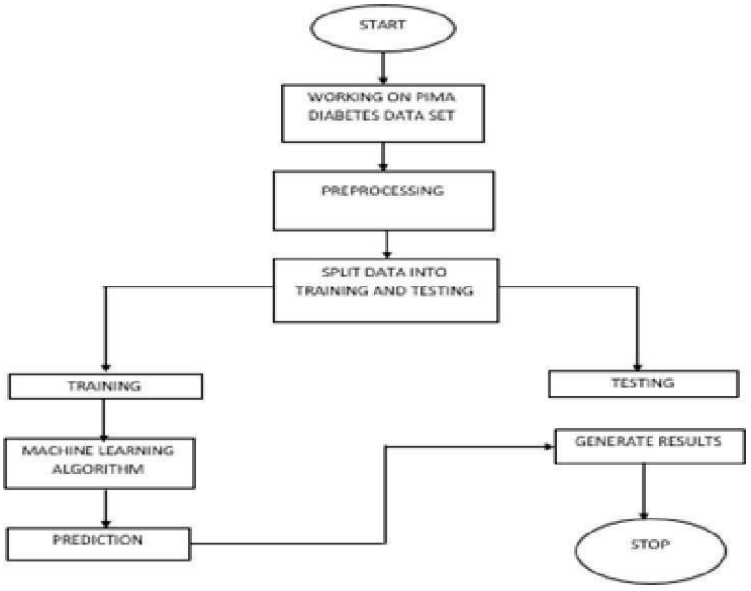

Figure 4.1: Proposed Block diagram

1. Data Collection:

The first step in the process involves gathering the necessary data for the diabetes prediction model. In this case, the Pima Diabetes dataset is used. This data set contains relevant information about individuals, such as age, BMI, and glucose levels, which can be used as features for predicting diabetes.

2. Preprocessing:

Before feeding the data into machine learning algorithms, preprocessing is crucial. This step involves tasks such as handling missing data, normalizing or scaling features, and dealing with outliers to ensure that the dataset is suitable for training and testing.

3. Splitting the Dataset for Training and Testing:

To assess the model's performance, the dataset is split into two parts: a training set and a test set. Typically,70% of the data is allocated for training purposes, and the remaining 30% is used for testing. The training set is employed to instruct the machine learning models and the test set is used to evaluate their ability to make accurate predictions.

4. Training with Machine Learning Algorithms:

In the training phase, various machine learning algorithms are employed to learn patterns

relationships within the dataset. Common algorithms for diabetes prediction include Support Vector Machine (SVM), Artificial Neural Network (ANN), Decision Trees, and Logistic Regression. The algorithms use the training data to adjust their parameters and build predictive models.

5. Testing: Generate Results:

After the machine learning models are trained, they are tested using the separate testing dataset. The models make predictions based on the features of the individuals in the testing set. These predictions are then compared to the actual outcomes (diabetes status) in the testing data to evaluate the models' performance

6. Output: Person is Affected with Diabetes or Not:

The final output of the process is a determination of whether an individual is affected by diabetes or not, based on the predictions made by the trained machine learning models. This output provides valuable information for early diagnosis and intervention, helping individuals take proactive steps towards managing their health.

7. Stop:

The process concludes after generating predictions for the testing dataset, with the results indicating whether the person is predicted to have diabetes or not. The model's accuracy and performance can then be assessed and further refined if necessary to improve its predictive capabilities.

Chapter 7

SNAPSHOTS

7.1 Project review.

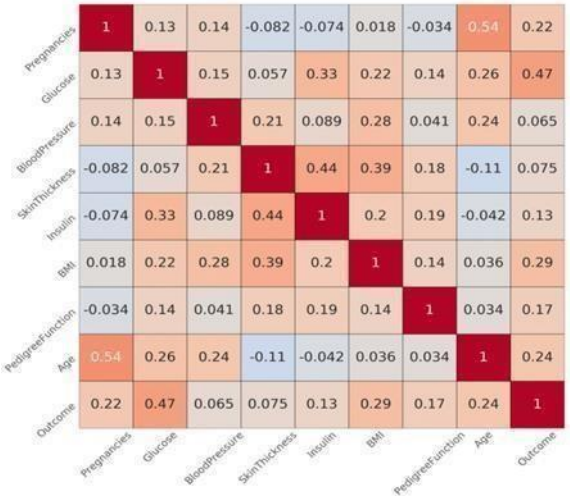

Figure 7.1 Heat map of Person's correlation coefficients for all diabetes features.

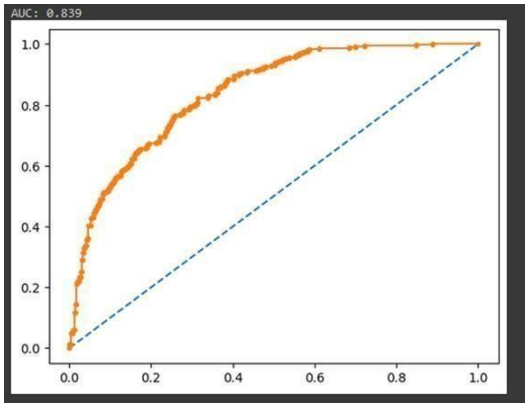

Figure 7.2 Logistic Regression

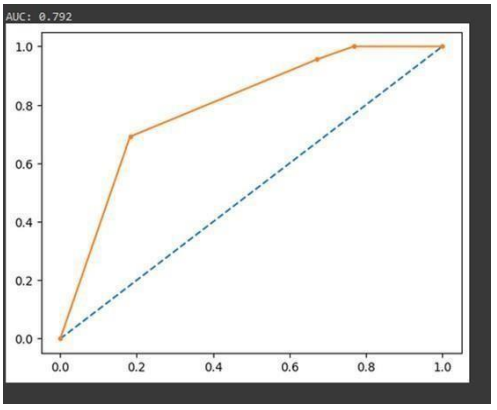

Figure 7.3 Decision Tree

Chapter 8

CONCLUSION

In conclusion, this internship has been a very useful experience for me. I can surely say that my understanding of the software environment has increased greatly. However, I do think that there are some aspects of the job skills that I could have done better and that I need to work on. I have built more confidence in usage of software tools like idle, google collab. The two main things I learnt after my experience in this firm are time management and being self-motivated. I have gained new knowledge and skills and met new people. Usage of python inbuilt function tools can improve operational efficiency. Usage of charts, maps, other visual representations of data to help present your findings in an easy way to understand. Improving the data visualization skill soften means learning visualization software. I came to know the real time application of python in social aspects. Finally overall experience is good.

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